# Ethan Labbe

Robotics engineer | Canada
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## EXPERIENCE

**Controls Engineer Intern | Tesla** Jan 2025 - Aug 2025

- Developed an automated vision-inspection station using 2D line-scan images and 3D profilometer point clouds to validate component geometry.
- Calibrated, tested, and validated vision systems to assess component features and identify dimensional faults.
- Programmed PLC motion control for industrial robots, servo motors, conveyors, and a dial table.
- Integrated PLC and industrial-PC vision systems over TCP/IP; configured Siemens TIA Portal HMIs, safety-rated I/O, and fail-safe PLC logic.

**Controls Engineer Intern | General Fusion** Jan 2024 - Aug 2024

- Designed and deployed a Python data-acquisition system for fusion diagnostics.
- Developed Python instrument drivers to control oscilloscopes, digitizers, and cameras.
- Integrated LabVIEW interfaces, DAQ devices, and TwinCAT Structured Text PLC code over TCP/IP; maintained modular code with Git.

**Controls Systems Integrator | Cybertech Automation** May 2023 - Sep 2023

- Led the design, installation, and documentation of a three-phase motor-control-center power-monitoring system.
- Designed a control subpanel to capture and communicate energy data over Modbus TCP.
- Configured, tested, and deployed custom SCADA/HMI interfaces.

## PROJECTS

**ROS 2 Edge Computing for Robotic Inspection** Jun 2026 - Present

- Developing a Raspberry Pi 4 inspection pipeline that offloads workloads when anomalies or resource limits are detected.
- Building a Gazebo hardware-in-the-loop environment to evaluate inspection performance with camera hardware.

**Handheld Stereo Vision Depth Mapping** Sep 2025 - Apr 2026

- Designed and built a handheld system combining a Raspberry Pi 5, AI cameras, a touchscreen, and an external viewer.
- Developed a Python/OpenCV pipeline for camera acquisition, stereo calibration, depth mapping, and wireless image transfer; validated depth accuracy to 2.5 mm.

**Inertial Fusion Fuel Target Research** Sep 2023 - Dec 2023

- Prepared laser-scattering experiments at the University of Alberta Laser Laboratory to evaluate inertial-fusion fuel targets.
- Captured and analyzed camera-based measurements to support fuel-target characterization and manufacturing-cost reduction.

## TECHNICAL SKILLS

**Programming:** Python, C/C++, MATLAB, Structured Text, Ladder Logic; Linux, Docker, Git/GitHub, CI/CD.

**Vision and robotics:** ROS 2, Gazebo, OpenCV, HALCON, camera calibration, line-scan and area-scan imaging, 3D point-cloud processing, sensor fusion.

**Controls and hardware:** Siemens TIA Portal, TwinCAT, PLCs, HMI/SCADA, safety I/O, TCP/IP, Modbus TCP, DAQ, KiCad, AutoCAD, circuit design.

## EDUCATION

**BSc, Electrical Engineering (Co-op) | University of Alberta** Class of 2026

Cumulative GPA: 3.8/4.0